brown open silicon

designing open chips. together.

October 18, 2026

Meet the Team



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Logistics

General Body Meetings
ERC 125
Sundays 4-5pm

In-Project Team Meetings
TBD

Questions? Contact: exiquio_salinas@brown.edu

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Icebreaker

If you had to stay in one building at Brown for the rest of your life, which would it be? (no dorms or dining halls)





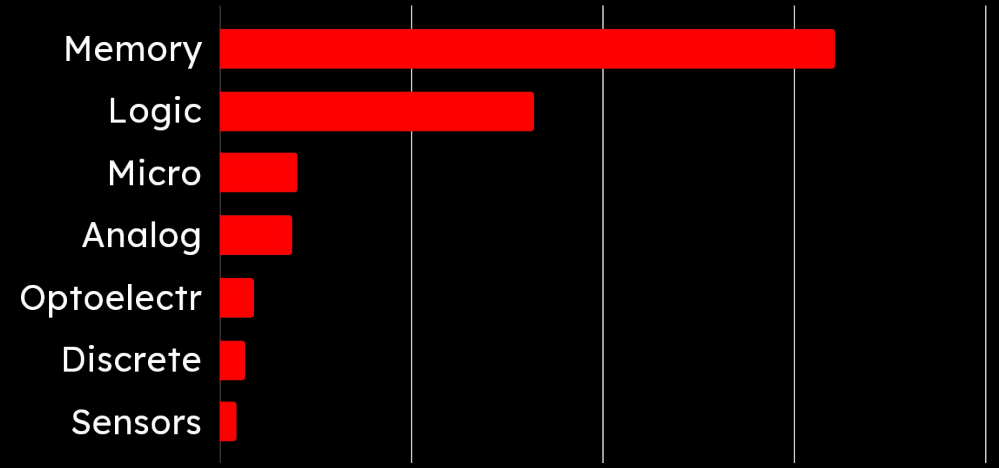
Semiconductor Industry

Global Chip Sales

2025 **$792B** +25.6% YoY

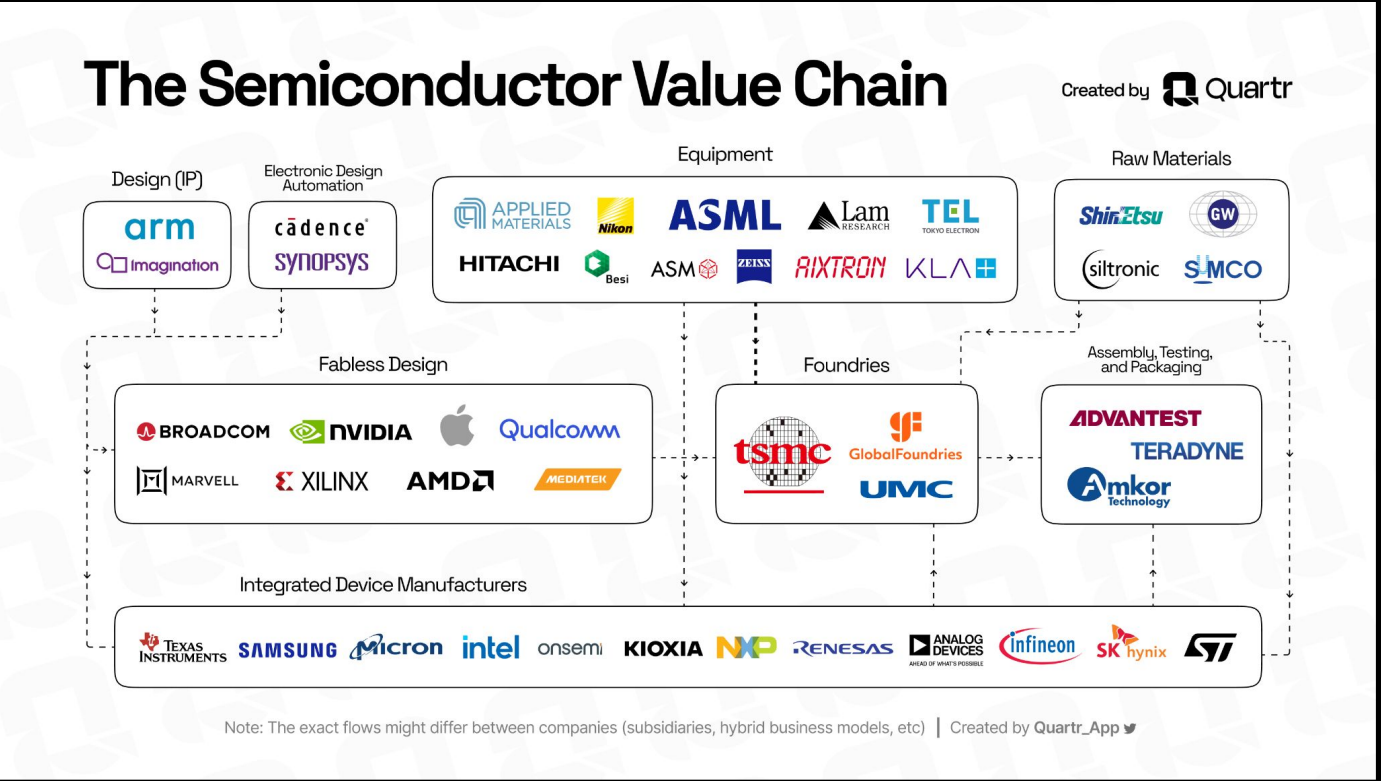
2026 (Forecasted) **$1.51T** +90% YoY

Industry Breakdown by Revenue



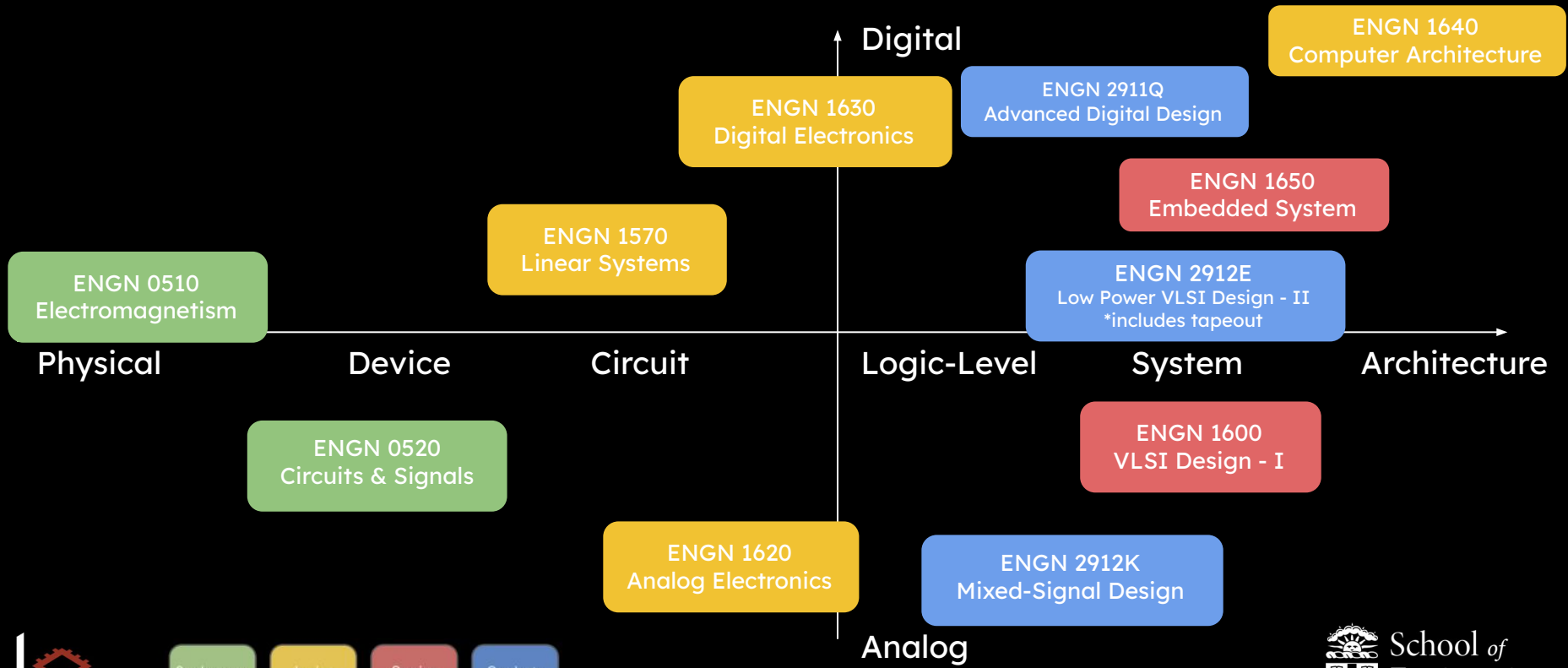
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Industry Companies



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ECE Classes at Brown



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Chip Design Process

1. Specification and Architecture

- Requirements that the chip must fulfill
  - how fast, how much power, what area and cost
- High-level structure
  - Cores, memory hierarchy, blocks
- Modeling tools: SystemC, C/C++, Python, MATLAB/Simulink

2. Design entry

- Capture the design in a form tools can work with





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Chip Design Process

3. Verification

- Prove the design works before any layout effort
- Writing testbenches and running simulations

4. Implementation

- Turn the design into real circuit elements
- A gate-level netlist (digital) or a finalized schematic (custom)





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Chip Design Process

5. Physical Design + Verification

- Create the actual geometry that will be manufactured
- Netlist turned into actual layout
  - Floorplanning, Placement, Routing
- DRC (design rule checks) and LVS (layout v. sch)

6. Signoff + Tape-out

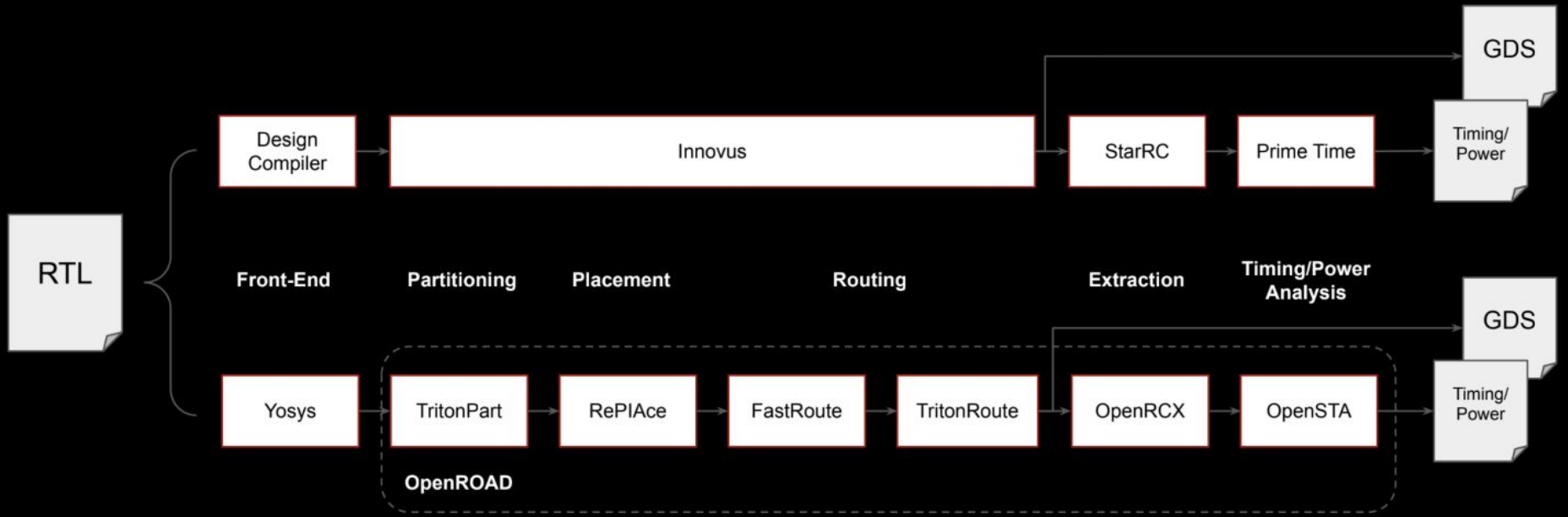
- Power and timing analysis
- Manufacturing the chip



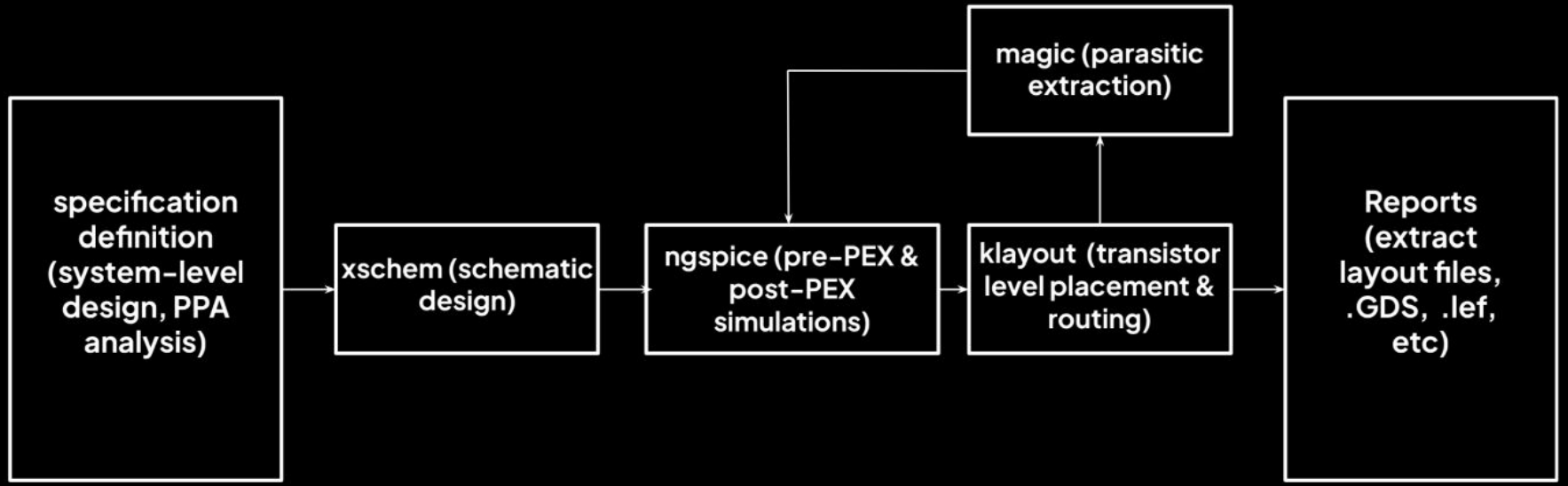


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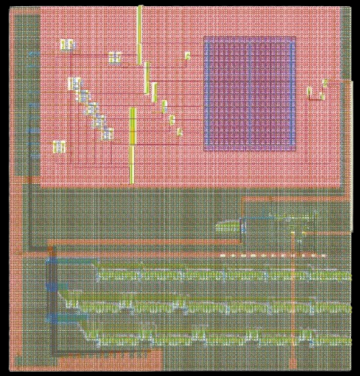
Digital Design Flow



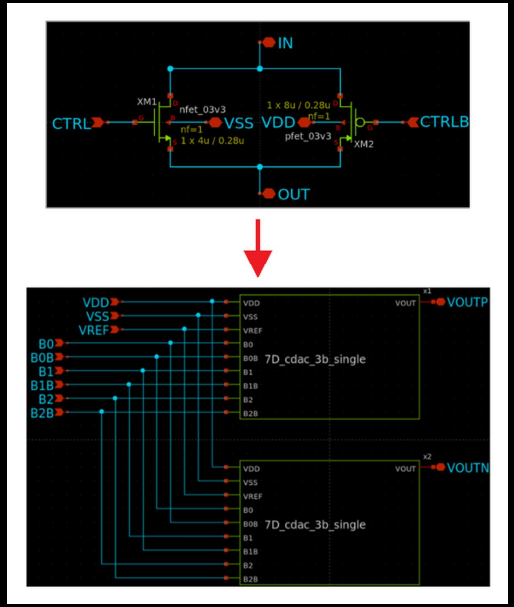
Custom/Analog Design Flow



Past Projects



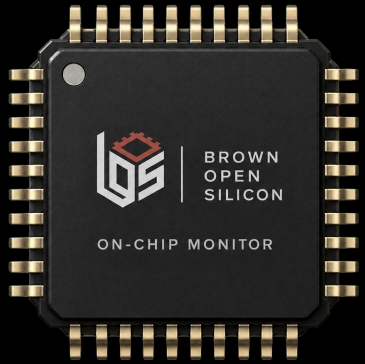
Entry into the 2026 Chipathon:
8-bit SAR ADC in gf180mcu PDK



Track D Project

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Introductory Project



On-chip Monitor

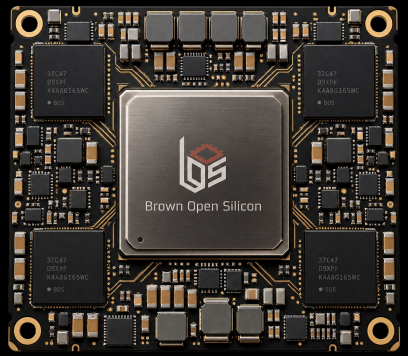
Timespan
1 year

Target Parameters
Measurement Accuracy, Response Time, Power Consumption

Overview
Measures temperature, circuit timing and variation, transistor aging

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Flagship Project



Accelerator for 8-bit Quantized Models

Timespan
3 Years

Target Parameters
Energy Efficiency, Latency, Compute Throughput, Area

Overview
Speed up neural network math through an 8-bit inference accelerator

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Team Options

1. Architecture

System Architecture, Compute & Memory Design, Performance Optimization

2. RTL Design

- SystemVerilog Development, Datapath & Control Logic, FPGA Prototyping

3. Design Verification

- Functional Simulation, Testbench Development, Hardware Debugging

4. Physical Design

- Place-and-Route, Timing & Physical Sign-off

5. Inference

- Software & Kernel Development, Performance Benchmarking





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This Week's Challenge

Analog

- Install Xschem + ngspice
- https://github.com/sscs-ose/sscs-chipathon-2026/blob/main/docs/install_instructions/Windows/install_docker_desktop.md
- Draw and simulate a CMOS inverter
- https://github.com/sscs-ose/sscs-chipathon-2026/tree/main/resources/Analog/eda

Digital

- Learn Verilog: https://hdlbits.01xz.net/wiki/Main_Page
  - Sections: Getting Started, Verilog Language: basic gates, vectors

Both

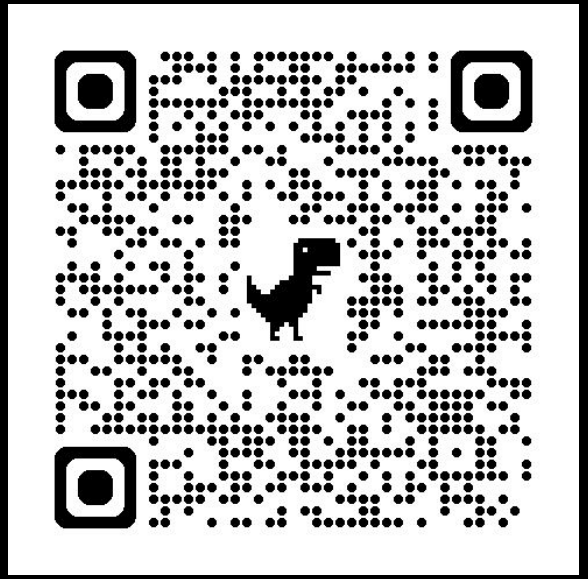
Review and familiarize yourself with design process in slideshow!

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Slack



Attendance



Website: https://brownopensilicon.org/team
Instagram: @brownopensilicon

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